Chapter 6

Fuel and combustion

Fuel

Any substance consists principally of carbon and hydrogen (hydrocarbons) and easily burning with air and give sufficient amount of energy.

Fuel composed mainly of

95% of fuel is carbon and hydrogen.

5% of fuel is :

- *metals(Fe, Pb, …),*
- *sulfur,*
- *gases(O_2, CO_2…)*
- *impurities.*

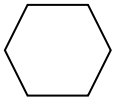
Hydrocarbons in fuel

Paraffin's

it is saturated compounds, having **high heat value** such as methane and ethane.

C–C–C–C–C

Naphthenes (alicyclic compounds (e.g.,cyclohexane)



Olefins it is Unsaturatedcompounds,eg. ethylene .

(-C=C-)

They have low heat value

Aromatic

it is cyclic unsaturated compounds, having low heat value such as benzene.

Classification of fuel

Solid fuel

as coal, coke……

Liquid fuel

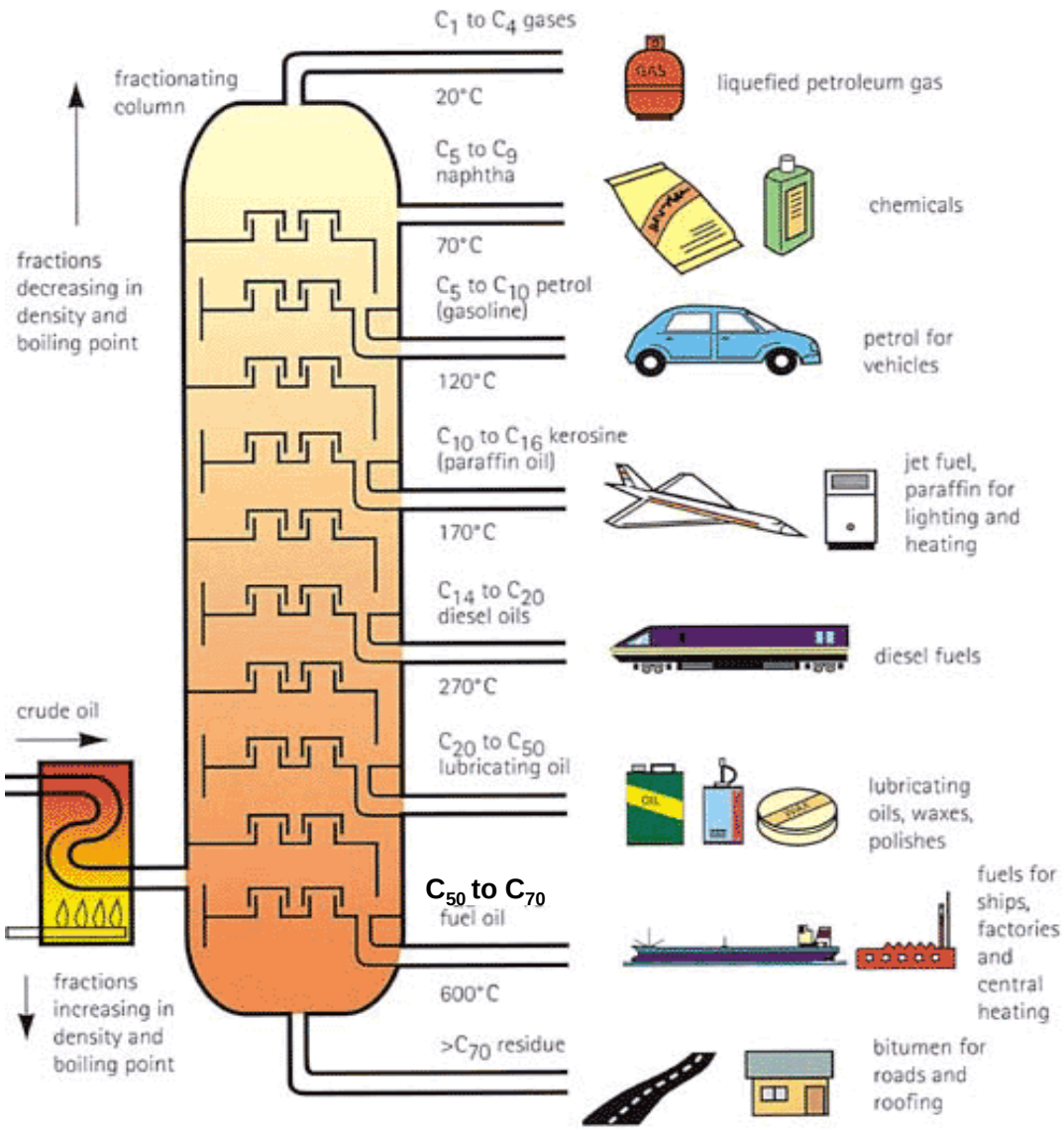
as gasoline, kerosene…..

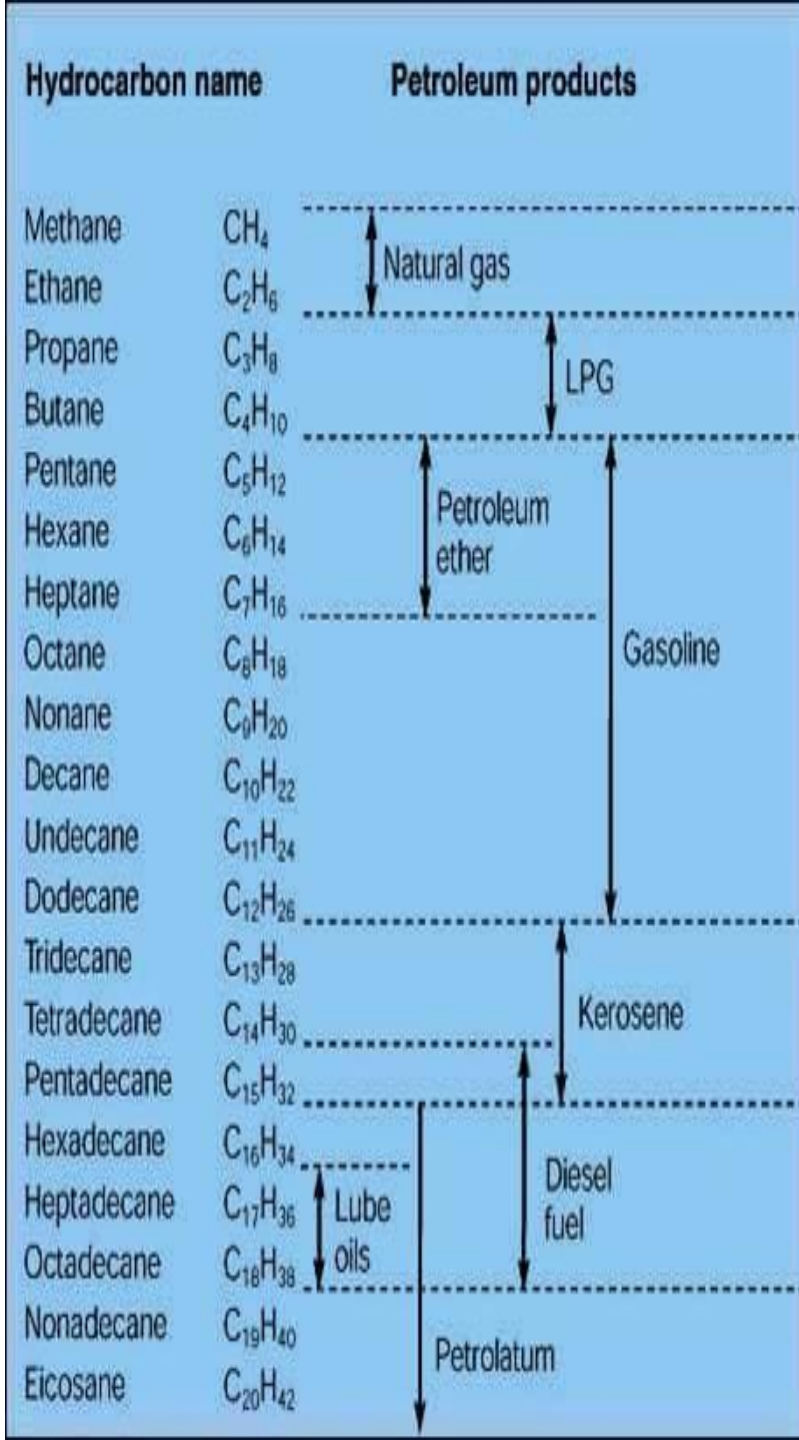
Gaseous liquid

as liquefied natural gas (LNG) and liquefied petroleum gas (LPG)

LNG Tanker

Petroleum refining processes





Properties of fuel

1. *Specific gravity*
2. *Viscosity*
3. *Calorific value*
4. *Surface tension*
5. *Pour point*
6. *Flash point*
7. *Spontaneous ignition temperature*
8. *Carbon to hydrogen ratio*
9. *Sulfur content*
10. *Ash content*

Properties of fuel

1. Specific gravity

Specific Gravity = relative density

- **It is the ratio between the density of the fuel at a specific temperature to the density of water at the same temperature.**
- **OR**
- **It is the ratio between the weight of a given volume of the fuel at a specific temperature to the weight of same volume of water at the same temperature.**

$$sp.gr. = \frac{\text{density of the fuel}}{\text{density of the water}} = \frac{\text{weight of a given volume of the fuel}}{\text{weight of a given volume of the water}} \quad \text{at specific temp.}$$

the sp.gr is a measure of the percent of paraffin's

- **sp.gr** ↑, ∴ % paraffin's ↓, ∴ bad ignition (incomplete combustion).
- **sp.gr** ↓ , ∴ % paraffin's ↑, ∴ good ignition (complete combustion).

Properties of fuel

2. Viscosity

It's the resistance of fluid to flow (a measure of the internal friction)

- Liquids which flows freely is said to have low viscosity.
- Liquids which have sluggish flowing is said to have high viscosity.

Viscosity

Dynamic viscosity:

is the force existing when a layer of fluid of unit area is moving at a unit velocity relative to similar layer at a unit distance apart and parallel to the original layer.

If force is measured in dyne, viscosity will be in poise.

It is related to dynamic force

μ = (force/area)(distance/velocity)

= (Dyne.sec/cm^2) or poise

Kinematic viscosity:

is the dynamic viscosity divided by density of fluid

Kinematic viscosity (γ)= μ / ρ ;

ρ: is the density of fluid

Unit : cm^2/sec or stoke

Dyne= g.cm.s^{-2}

Viscosity ↑ ∴ % paraffin's ↓, ∴ bad ignition (incomplete combustion) (large particle size)

Viscosity ↓ ∴ % paraffin's ↑, ∴ good ignition (complete combustion) (small particle size)

Properties of fuel

3. Calorific value CV

It's the amount of heat evolved by burning the unit mass of fuel with oxygen.

the amount of *heat* released by a unit weight or unit volume of a substance during complete combustion

Unit: (MJ/kg) or (J/g) or (Cal/g) or (Btu/lb)

BTU or Btu (British Thermal Unit) is the amount of heat energy needed to raise the temperature of one pound of water by one degree F.

The heat of combustion for fuel is expressed as the HCV (GCV) or LCV (NCV).

HCV = Higher calorific value = gross CV

LCV = Lower calorific value = actual or net CV

Due to presence of hydrogen in fuel its burning is associated with water vapor formation and decreasing the heat content

$$(C, H) + O_2 \rightarrow CO_2 + H_2O + \text{heat}$$

A compound composed of carbon and hydrogen reacts with the oxygen

to produce carbon dioxide, water and heat

Higher calorific value (HCV) or (GCV) is defined as the amount of heat evolved when a unit weight (or volume in the case of gaseous fuels) of the fuel is completely burnt and the products of combustion cooled to the normal conditions (with water vapor condensed as a result). The heat contained in the water vapor must be recovered in the condensation process.

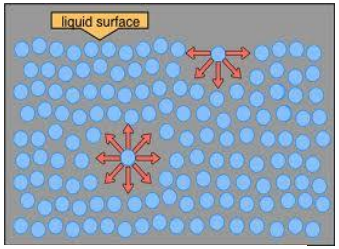
Lower calorific value (LCV) or (NCV) is defined as the amount of heat evolved when a unit weight (or volume in the case of gaseous fuels) of the fuel is completely burnt and water vapor leaves with the combustion products without being condensed.

HCV = LCV + Latent heat of condensation of water

The difference between LCV and HCV (or Lower and Higher Heating Value, or Net and Gross) is clearly understood by all energy engineers. There is **no** 'right' or 'wrong' definition. The numerical difference between the two is the latent heat of condensation of the water vapour in the combustion exhaust gas, which in turn depends on the hydrogen content of the fuel being burned. The difference is minimal for coal, significant for natural gas and largest for pure hydrogen fuel. So you have to do the conversion calculation for the specific case. Unfortunately, for historical reasons, the natural gas (public supply) and coal industry (inc' coal-fired power stations) have grown up using HCV, while engines (all fuels, inc' diesel) and gas turbines have ended up using LCV basis, giving slightly higher nominal % efficiencies for those products. This is not about 'salesmen and spin', it's just engineers using different bases.

Properties of fuel

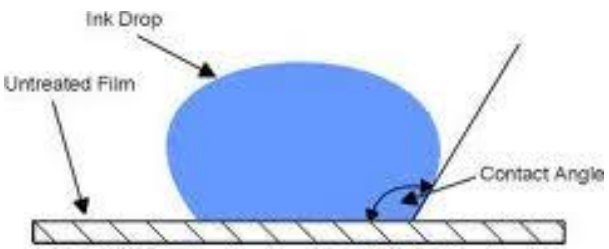
4. Surface tension



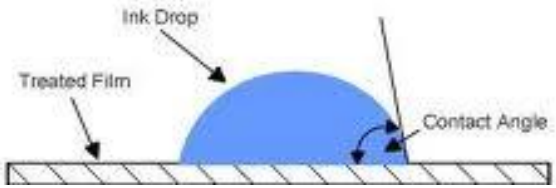
"The molecules inside the liquid are affected by attraction forces in all direction by other molecules (inside the liquid). However, molecules at the surface are pulled downward and sideways by other molecules, but not upward away from the surface. Thus, the surface layer will be in tension state and affected by surface tension forces". Thus the liquid's surface try to decrease its area to decrease this surface tension (by taking the ball or the drop shape).

"It is the work (erg. cm^{-2}) or energy required to stretch or increase the surface area of liquid by 1 cm^2 "

Intermolecular forces ↑, surface tension increases ↑



- Higher surface tension (cohesive force higher than adhesive force) lead to bad ignition



- lower surface tension (cohesive force lower than adhesive force) lead to good ignition.

Properties of fuel

5. Pour point

The **pour point** of a liquid is the highest temperature at which it becomes semi solid and loses its flow characteristics

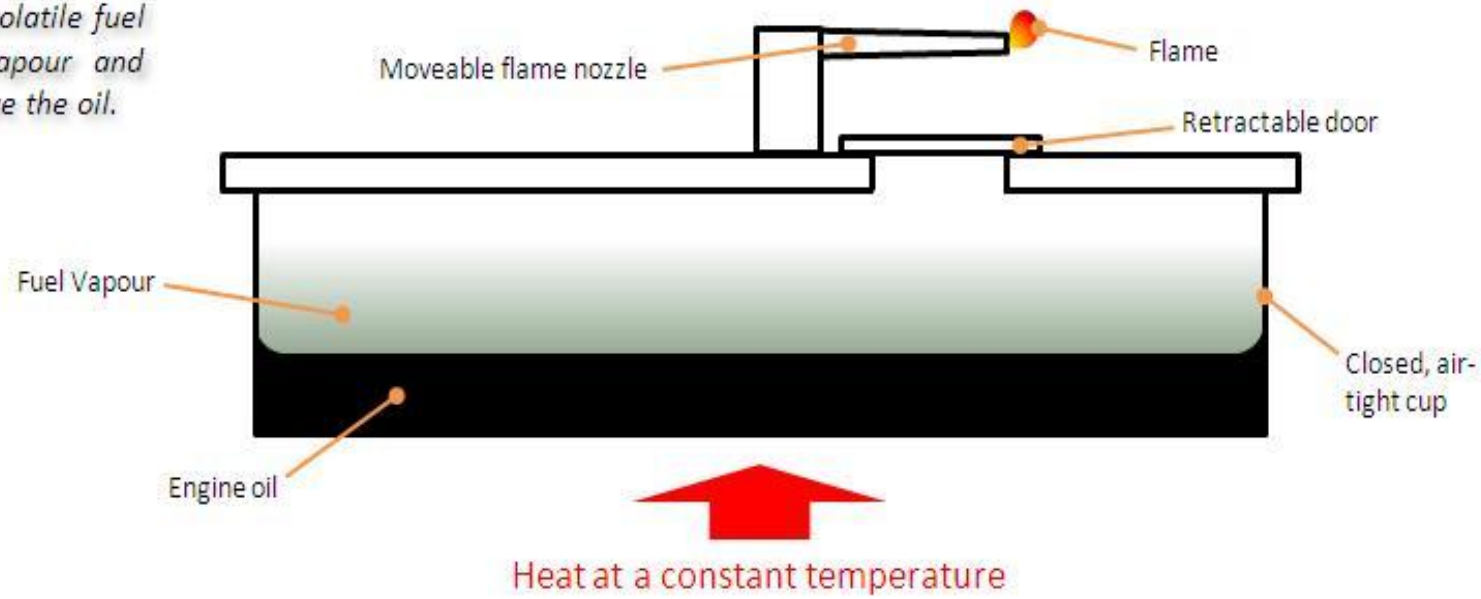
it can become the determining factor in selecting one lubricant from among a group with otherwise identical properties.

6. Flash point (F.P)

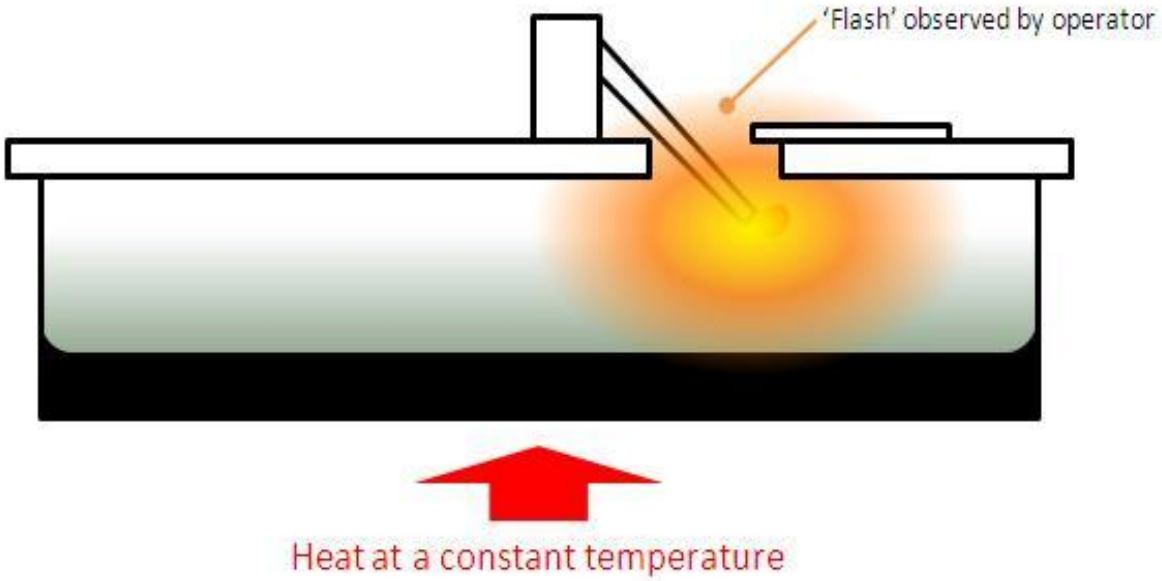
- The temperature at which the fuel oil will form enough amount of flammable mixture to ignite with air in presence of flame.
- Flash point give good indication for volatility of fuel, storage condition, safety
- Flash point is inversely proportional with the fuel volatility.
- Gasoline(F.P)= 22 C°
- Diesel oil(F.P)= 65 C°
- Low flash point → volatile fuel → good ignition → High CV

Flash Point Test

The oil is heated at a constant temperature for a set time.
During this time, any volatile fuel will evaporate into vapour and collect in the space above the oil.



After this time, the flame is introduced into the closed cup and a flash will be seen if enough fuel vapour has collected.
The test will be reported as 'negative' if no flash is seen.



7. Spontaneous ignition temperature (S.I.T)

- The temperature at which the fuel starts to ignite with air without flame or spark.
- S.I.T is much greater than F.P.
- Gasoline(F.P)= 22 C°
- Gasoline(S.I.T) = 260 C°

8. Carbon to hydrogen ratio

- An increase of the carbon than hydrogen associated with decrease the percent of paraffin's and increase Sp.gr, viscosity and decreasing the C.V which tends to bad ignition.

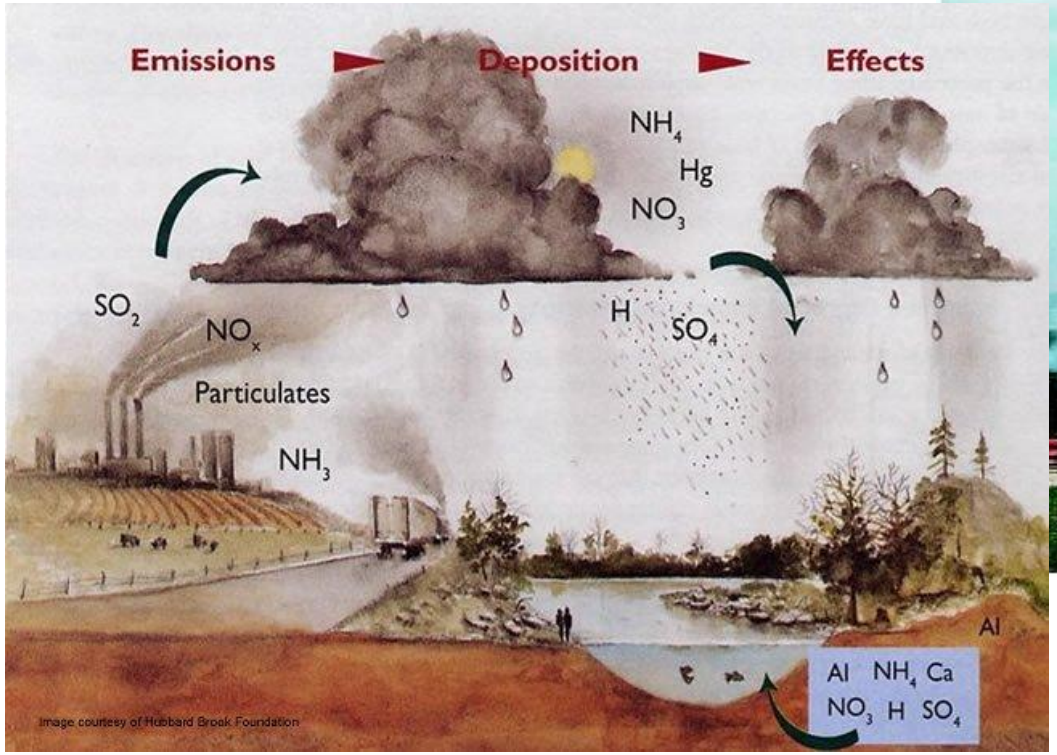
C–C–C–C–C

- Higher percent of carbon give black smoke (C) during ignition.
- (C/H) ↑, ∴ Oil quality & price ↓.

9. Sulfur content

- Sulfur in fuel refers to free sulfur and its compounds such as thioether CH_3-S-CH_3, mercaptens CH_3-SH, hydrogen sulfide H_2S.
- Presence of sulfur in fuel associated with formation of sulfur oxide gases (which react with water or dew (الندى) forming sulfuric acid and tends to air pollution and formation of strong corrosive environment.
- $S + O_2 \rightarrow SO_3 \xrightarrow{H_2O} H_2SO_4$
- S ↑, ∴Oil quality & price ↓.

Air Pollution



10. Ash content

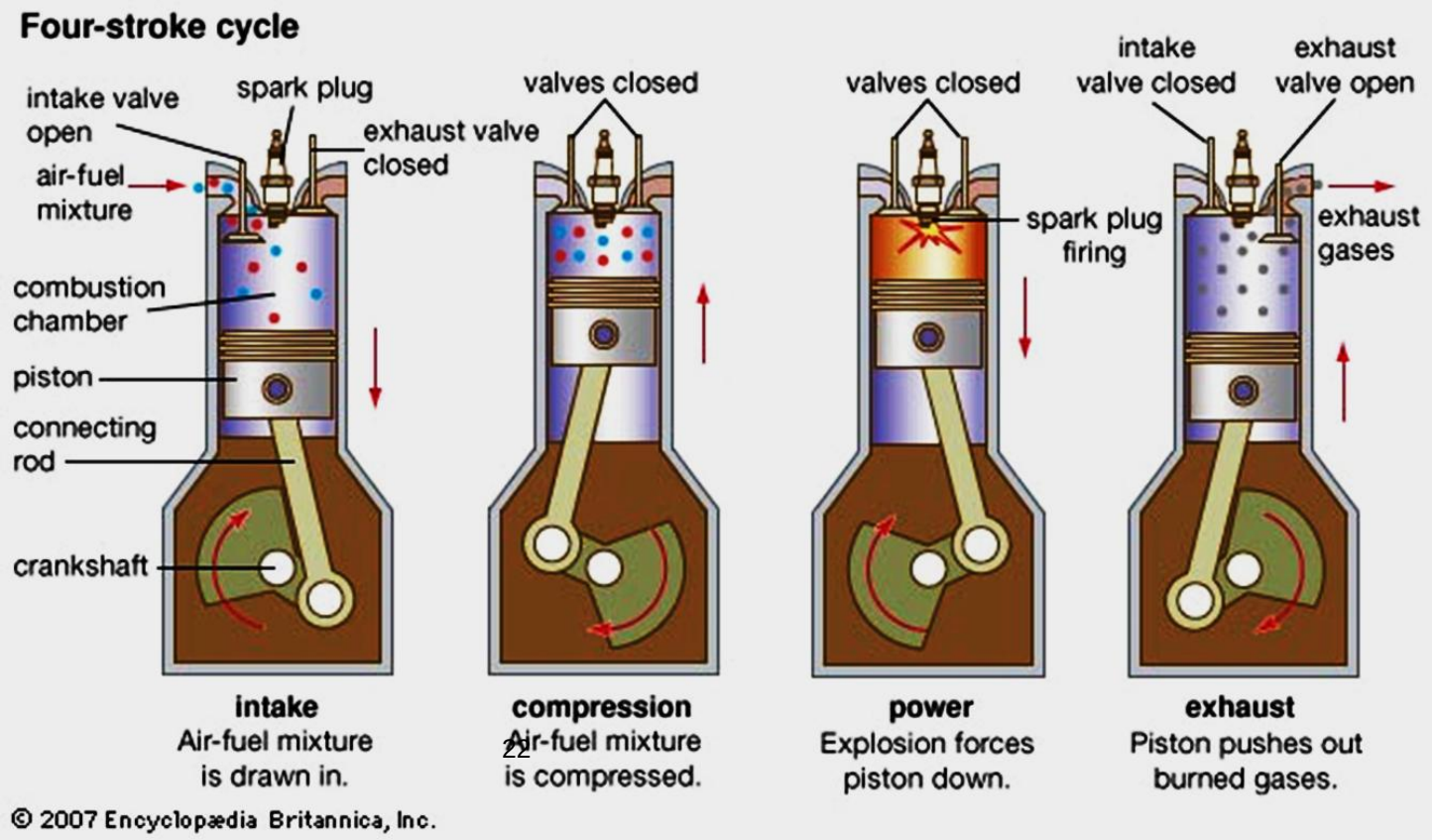
- Refers to non combustible material present in fuel.
- Ash in fuel present as all inorganic compound such as metals(Al, Ca, Fe, Mg, Ni, V, Na, …) , metal oxides and inorganic salts.
- Percent of ash in normal liquid fuel is about 0.1%.
- Separation of ash is uneconomically process.
- % Ash ↑, ∴Oil quality & price ↓

Parameter	Good fuel	Bad fuel
	good ignition complete combustion small particle size	bad ignition incomplete combustion Large particle size
HCV (LCV)	↑	↓
% Paraffin	↑	↓
C/H	↓	↑
Sp. gr.	↓	↑
Viscosity	↓	↑
Surface Tension	↓	↑
Pour point	X or ↓↓	↓
F.P. or (1/Volatility)	↓	↑
S.I.T (>> F.P)	↓	↑
Sulfur content	↓	↑
Ash%	↓	↑

II- Fuel & Combustion

- In combustion reactions, rapid oxidation of combustible elements of fuel occur and produce combustion products and release energy.
- The three major combustible chemical elements in most common fuels are C, H, and S.
- Although sulfur is usually a relatively unimportant contributor to the energy released, it can be a significant cause of pollution and corrosion.

Internal Combustion Engine



Combustion Reactions

1. Complete combustion of carbon:

$$C + O_2 \rightarrow CO_2 + \text{Heat (33.7 Mega joule/kg).}$$

2. Incomplete combustion of carbon :

$$C + 1/2\ O_2 \rightarrow CO + \text{Heat (10.5 Mega joule/kg).}$$

3. Hydrogen Combustion:

$$H_2 + 1/2O_2 \rightarrow H_2O + \text{Heat (65.4 Mega joule/kg).}$$

4. Sulfur combustion:

$$S + O_2 \rightarrow SO_2 + \text{Heat (9.11 Mega joule/kg).}$$

Deduce the complete combustion equation of Carbon ?

$C + O_2 \rightarrow CO_2$ + Heat (33.7 Mega joule/kg).

1mole	1mole	1mole
(At.wt*1)	(M.wt*1)	(M.wt*1)
(1*12)	(1*2*16)	(1*(12+2*16))
12kg	32kg	44kg

Divide all masses by 12 (the smallest value):

1kg	8/3 kg	11/3 kg

1kg C + 8/3 kg $O_2 \rightarrow$ 11/3 kg CO_2 + (33.7 MJ/kg)

^{1}H
^{12}C
^{14}N
^{16}O
^{32}S

Deduce the incomplete combustion equation of Carbon ?

$C + \frac{1}{2} O_2 \rightarrow CO$ + Heat (10.5 Mega joule/kg)

1mole	½ mole	1mole
12kg	16kg	28kg

Divide all masses by 12:

1kg	4/3 kg	7/3 kg

1kg C + 4/3 kg $O_2 \rightarrow$ 7/3 kg CO + (10.5 MJ/kg)

Deduce the combustion equation of Hydrogen ?

H_2 + ½ O_2 → H_2O + Heat (65.4 MJ/kg).

1mole ½ mole 1mole

2 kg 16 kg 18 kg

Divide all masses by 2:

1kg 8 kg 9 kg

1kg H_2 + 8 kg O_2 → 9 kg H_2O + (65.4 MJ/kg)

Deduce the combustion equation of Sulfur ?

S + O_2 → SO_2 + Heat(9.11Mega joule/kg).

1 mole 1 mole 1 mole

32kg 32 kg 64kg

Divide all masses by 32:

1kg 1 kg 2 kg

1kg S + 1 kg O_2 → 2 kg SO_2 + (9.11 MJ/kg)

^{1}H
^{12}C
^{14}N
^{16}O
^{32}S

Comparison between combustion of carbon, hydrogen, sulfur ?

	Carbon complete	Carbon incomplete	Hydrogen	Sulfur
Combustion reactions				
Amount of oxygen needed to burn 1 kg of the combustible component	8/3 kg	4/3 kg	8 kg	1 kg
Amount of heat released	33.7 MJ	10.5 MJ	65.4MJ	9.11 MJ
combustion Products	CO_2	CO	H_2O	SO_2
Environmental impact	pollutant	Toxic and pollutant	No impact	Pollutant and corrosive.

Deduce the combustion equation of methane ?

CH_4 + 2 O_2 → CO_2 + 2 H_2O + Heat

16 kg 64 kg 44 kg 36 kg

Divide all masses by 16:

1kg 4 kg 11/4 kg 9/4 kg

1kg CH_4 + 4 kg O_2 → 11/4 kg CO_2 + 9/4 kg H_2O + Heat

^{1}H
^{12}C
^{14}N
^{16}O
^{32}S

Deduce the combustion equation of ethane?

C_2H_6 + 7/2 O_2 → $2CO_2$ + 3 H_2O + Heat

((2*12)+(6*1)) ((7/2)*2*16) (2*(12+(2*16))) (3*((2*1)+16))

30kg 112kg 88kg 54kg

Divide all masses by 30:

1kg 3.73 kg 2.93 kg 9/5 kg

1kg C_2H_6 + 3.73 kg O_2 → 2.93 kg CO_2 + 9/5 kg H2O + Heat

Deduce the combustion equation of propane?

C_3H_8 + 5 O_2 → $3CO_2$ + 4 H_2O + Heat

44kg 160kg 132kg 72kg

Divide all masses by 44:

1kg 3.63 kg 3 kg 1.6 kg

1kg C_3H_8 + 3.63 kg O_2 → 3 kg CO_2 + 1.6 kg H2O + Heat

Deduce the combustion equations of methanol, ethanol & propanol ?

^{1}H
^{12}C
^{14}N
^{16}O
^{32}S

Deduce the combustion equation of methanol ?

CH_3OH + 3/2 O_2 → CO_2 + 2 H_2O + Heat

32 kg 48kg 44 kg 36 kg

Divide all masses by 32:

1kg 3/2kg 11/2 kg 9/2 kg

1kg CH_3OH + 3/2 kg O_2 → 11/2 kg CO_2 + 9/2 kg H_2O + Heat

Deduce the combustion equation of ethanol?

C_2H_5OH + 3 O_2 → $2CO_2$ + 3 H_2O + Heat

((2*12)+6+16) ((3)*2*16) (2*(12+(2*16))) (3*((2*1)+16))

46kg 96kg 88kg 54kg

Divide all masses by 46:

1kg 2.09 kg 1.91kg 1.17kg

1kg C_2H_5OH+ 2.09 kg O_2 → 1.91 kg CO_2 + 1.17 kg H2O + Heat

Air supply

Air supply = amount of air added to combustion engine to burn the fuel inside the engine.

(Air composition = 78.1% **v/v** N_2 + 20.9% **v/v** O_2 + ~ 1% other constituents)

(Air composition = 75.5%wt/wt N_2 + **23.2% wt/wt O_2** + ~ 1% other constituents)

Air supply = total amount of oxygen * (100/23)

Air supply = [O_2 **for C(complete)** + O_2 **for C(incomplete)** + O_2 **for H** + O_2 **for S** - O **in Fuel**] * (100/23). (**and ignoring the reaction between N_2 and O_2**)

Amount of O_2 = (% of element * conversion factor * wt. of the fuel)

Air supply = [(%**C** * $\%_{complete}$ * 8/3) + (%**C** * $\%_{incomp}$ * 4/3) + (%**H*** 8) + (%**S** * 1) – (% **O** $_{in\ fuel}$)] *(wt. $_{fuel}$) * (100/23)

(**and ignoring the reaction between N_2 and O_2**)

Deduce the air supply for 70% complete combustion of 2kg methanol ?

CH_3OH = 12 + 4x1 + 16 =32

% C = 1x12/32 = 37.5 %

% H = 4x1/32 = 12.5 %

% O = 1x16/32 = 50 %

Air Supply = [(37.5/100)*8/3*(70/100) + (37.5/100)*4/3*(30/100) + (12.5/100)*8 -(50/100)] * 2 * (100/23) = 4.13 kg Air

Excess Air

Theoretical Air. (1)
Actual air added (1+X)

X = (% of excess air)

Actual air added = (1+X)*Theoretical Air.

Example: calculate the actual air needed for the combustion of a fuel if the actual supplied air exceeds the theoretical air supply by 40% (and Th. air is 5 kg)

Actual air = (1+X)*Theoretical Air.

Actual air = (1+ (40/100))*5 =7 kg

Example: Calculate the Actual Air supply needed to burn 2 kg of the fuel which consists of : 60%wt.$_{carbon}$, 20%wt.$_{hydrogen}$, 5%wt.$_{sulfur,}$ 5%wt.$_{oxygen,}$ 9%wt.$_{water}$, 1%wt.$_{ashes.}$ **If the combustion is 90% complete & If the percentage of excess air is 40%.**

Air supply = [$\%_C$ * $\%_{complete}$ * 8/3) + ($\%_C$ * $\%_{incomp}$ * 4/3) + ($\%_H$* 8) + ($\%_S$ * 1) – (% $_{O\ in\ fuel}$)] *(wt. $_{fuel}$) * (100/23)

Theo. Air supply = [(0.6*0.9*8/3) + (0.6*0.1*4/3) + (0.2* 8) + (0.05*1) – (0.05)] * 2 *(100/23)

Theo. Air suppl. = [1.44 + 0.08 + 1.6 + 0.05 – 0.05]*2*(100/23) = 27.13 kg air

& If the percentage of excess air is 40% of the theoretical air supply.

Actual air supply = (1+X)*Theoretical Air.

Actual air supply = (1+0.4)*27.13= 37.98 kg of air.

Energy released by burning the fuel:

- Energy released = amount of energy produced due to combustion of the fuel inside the engine (its unit is Mega (or million) joule).
- Total energy = $[(E)_{C(complete)} + (E)_{C(incomplete)} + (E)_{H} + (E)_{S}] * (wt._{fuel})$

$$\text{Energy} = [(33.7 * \%C * \%_{complete}) + (10.5 * \%C * \%_{incomplete}) + (65.4 * \%H) + (9.11 * \%S)] * (wt._{fuel}) \quad MJ$$

Calculate the heat energy released during burning of 2 kg of the fuel which consists of : 60%wt. $_{carbon}$, 20%wt. $_{hydrogen}$, 5%wt. $_{sulfur,}$ 5%wt. $_{oxygen,}$ 9%wt. $_{water}$, 1%wt. $_{ashes.}$ **If the combustion is 90% complete?**

$$\text{Energy} = [(33.7 * \%C * \%_{comp}) + (10.5 * \%C * \%_{incomp}) + (65.4 * \%H) + (9.11 * \%S)] * (wt._{fuel})$$

$$\text{Energy} = [(33.7*0.6*0.9) + (10.5*0.6*0.10) + (65.4*0.2) + (9.11*0.05)] * 2$$

$$= 32.364 \text{ MJ}$$

Example: calculate the air needed to burn 3 kg of fuel of (50% methane, 50% propane). If the excess of air is 30% of the theoretical air.

Oxygen needed to burn the methane:

CH_4	+	$2\ O_2$	$\rightarrow$	CO_2	+	$2\ H_2O$
1mole		2 mole		1mole		2 mole
16kg		64kg		44kg		36kg

Divide all masses by 16:

1kg CH4 + 4 kg O_2 $\rightarrow$ 11/4 kg CO_2 + 9/4 kg H2O

1kg	4 kg	11/4 kg	9/4 kg
(3*50%))kg	**?** kg		

$\therefore$ **Oxygen needed to burn the methane in mix. = 3*0.5*4/1= 6 kg**

Oxygen needed to burn the propane:

C_3H_8	+	$5\ O_2$	$\rightarrow$	$3CO_2$	+	$4\ H_2O$ + Heat
1mole		5 mole		3 mole		4 mole
44kg		160kg		132kg		72kg

Divide all masses by 44:

1kg C_3H_8 + 3.63 kg O_2 $\rightarrow$ 3 kg CO_2 + 1.6 kg H2O

1kg	3.63 kg	3 kg	1.6 kg
1.5 kg	**?** kg		

$\therefore$ **Oxygen needed to burn the propane =1.5*3.63/1= 5.445kg**

- Total amount of theoretical air = (6 + 5.445) * (100/23) = 49.76 kg of air.
- Total amount of actual air = 49.76*(1+ 0.3) = 64.69 kg of air.

Homework: calculate the air supply required to burn 5 kg of fuel of a Brazilian car (30% methanol, 70% ethanol)? If the excess of air is 40% of the theoretical air? Then calculate the heat energy released?

Answer key:

O_2 needed to burn CH_3OH in fuel = 5*0.3*1.5/1 = 2.25 kg
O_2 needed to burn C_2H_5OH in fuel =5*0.7*2.087/1= 7.3045kg

Total amount of theoretical air = (2.25 + 7.3045) * (100/23) = 41.5 kg of air.
Total amount of actual air = 41.5*(1+ 0.4) = 58.16 kg of air.

% C = [(0.3*12/32) + (0.7*24/46)]*100 = (0.1125 + 0.3652) *100 = 47.77%
% H = [(0.3*4/32) + (0.7*6/46)]*100 = (0.0375 + 0.0913) *100 = 12.88%
% O = [(0.3*16/32) + (0.7*16/46)]*100 = (0.15 + 0.2435) *100 = 39.35%

Energy = [(33.7 * %C * $\%_{complete}$) + (10.5 * %C * $\%_{incomplete}$) + (65.4 * %H) + (9.11 * %S)] * (wt. $_{fuel}$)
Energy = [(33.7 * 0.4777) + (65.4 * 0.1288)] * (5) = 122.61 MJ